

COVID-19 Visualization Platform Based on Population Density Propagation Model

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Abstract—Based on the classical SIR model and CEMM intercity model, a new model was established by adding "population density" parameter to analyze and predict the spread of virus. In addition, the current trend of the epidemic and forecast data can be referenced to the public in an intuitive web view to improve the perception of risk information in the society. The real-time epidemic data interface was adopted to analyze the real-time pneumonia epidemic data captured by the deployment of timing crawler combined with the regional population density to build a model. Then, the diversified charts, Python and Web front-end technologies were used to realize the visualization of epidemic information. COVID-19 grows exponentially without obstruction, and when a place has a high population density, the spread of the virus accelerates and the number of people infected increases. The research shows that the integration of population density parameters can further improve the epidemic prediction function, provide epidemic data reference in a more effective and accurate way, and further improve the public's ability to perceive social risk information.

Keywords—COVID-19, population density, echarts, visualization, prediction model

I. INTRODUCTION

With the advent of the information age, the amount of data is increasing rapidly. Data visualization technology can transform massive data into charts and effectively convey information by certain interactive means, so that people can intuitively understand the data and dig the value behind the data. As of March 15, 2022, the cumulative number of confirmed COVID-19 cases worldwide had exceeded 456.79 million, with over 6.04 million deaths. COVID-19 virus is a serious threat to human life. Despite the successful development of novel coronavirus vaccine, COVID-19 still has the characteristics of asymptomatic, highly contagious, diverse influencing factors, and complex transmission routes. It is particularly difficult to treat the spread of the epidemic from the root, so it is necessary to accurately predict the epidemic and provide intuitive data information. It is of great and far-reaching significance to increase public risk awareness.

Scientific research workers spread of infectious diseases is not derived from the mathematical model is often used to condition, forecast analysis epidemic trend with an infected person, etc., to specify the corresponding prevention and control measures accordingly. There are a variety of existing transmission prediction models, among which SIR model is the most classic. SIR model has simple structure and strong operability, and is widely used for describing the overall trend of disease development. For example, Yin Nan^[1] conducted a simulation analysis of novel coronavirus transmission in a limited area based on SIR model. Yu Zi et al.^[2] predicted the

inflexion point and the maximum number of confirmed cases by using the SIR model based on time-varying parameters. Changes in population density caused by vulnerable person S(Suspected) and infected person I(Infected) on the move are often overlooked, resulting in a mismatch with expected data. Population movement during holidays and rush hours is more likely to cause changes in population density per unit time in an area, and the increase of population density will be one of the driving factors of epidemic spread. Therefore, in the face of constantly mutating Novel Coronavirus strains, it is particularly important to integrate population density parameters into prevention and prediction.

In this paper, taking a province in China as an example, we study the relationship between its COVID-19 infection data and regional area size and population size, and analyze the impact of population density of multiple cities in the urban circle on the spread rate of the virus. The "population density" parameter is integrated into the transmission model, and the COVID-19 epidemic information is visualized. Through the acquisition of dynamic data, the regional epidemic development trend is predicted and displayed in real time, so as to provide effective epidemic prevention measures and travel advice.

II. RELATED WORK

A. SIR Model

The SIR model is adapted to study the infection situation in a closed area, and can accurately fit the curve according to the existing data, and use the phase rail line analysis to obtain measures to prevent the spread of infectious diseases, and the theoretical basis is sufficient. SIR model is one of the most widely used virus transmission models as a classical model to study the spread of infectious diseases, no matter in the research on prevention of smallpox virus or the black Death in meteor shower London.

However, the classification of population s in the SIR model is not detailed enough to explicitly consider population density, and the reality of the flow of crowd control is an effective means to prevent the spread of the epidemic. Second, there is a probability that people who are infected will still be infected after they recover, which leads to a bias in the predicted data. A minimalist model based on a small sample differs from a real large population. In the process of forecasting, it is inevitable to reduce the accuracy of forecasting data for a long period of time simply based on existing data. In particular, the change of population density caused by the uncertainty of population flow cannot reflect the influence of this factor on epidemic control by using existing data, which is a big omission of SIR model. Therefore, population density parameter must be introduced

to achieve more accurate prediction function. Based on the above considerations, we introduced "population density" as an important parameter.

For the population in the SIR model, researchers often divide the population into suspected (S), infected (I), and recovered (R). If a suspected person is within the contagious range of a patient, he is at risk of becoming infected. Assuming that the total population at time t is $N(t)$, then $N(t) = S(t) + I(t) + R(t)$. Assume that the number S of suspected people that each infected person I contacts daily is a constant λ , and the number of recovered people R is a constant μ proportional to the infected person. Then, the state transition differential equation and basic graphical representation of the SIR model can be obtained as Equations (1), (2), (3) and Figure 1.

$$\frac{ds}{dt} = -\lambda \cdot i(t) \cdot s(t) \quad (1)$$

$$\frac{di}{dt} = \lambda \cdot i(t) \cdot s(t) - \mu i(t) \quad (2)$$

$$\frac{dr}{dt} = \mu i(t) \quad (3)$$



Fig. 1. The path diagram of the SIR model state transition

B. Population Density

Population Density refers to the average number of Population on a given unit of land in a given period of time. Population density is a commonly used quantitative index to reflect the density of population distribution, which is usually used to calculate the population distribution of a certain country, region, city or the world. It's calculated by dividing its total population by its total area.

C. CEMM Model

CEMM think every city increased cases of infection is mainly caused by two main processes: from the spread of other cities and the spread of the inner cities. CEMM combined the intercity transmission and intra-city spread of infectious diseases to estimate the infection cases in each city. It is an efficient research model to analyze and study epidemic transmission from the perspective of urban network. It can be used to simulate the spatial transmission of novel Coronavirus.

CEMM model based on urban network perspective, emphasizing the city population flow and the important role of rapid transit network spread of epidemic diseases. Secondly, the model understands the spatio-temporal mechanism of epidemic transmission from the perspective of geographic network, which is conducive to the accurate formulation of corresponding prevention and control measures.

However, case diagnosis is a complex process, and the number of reported cases cannot fully reflect the actual number and status of infection. At the same time, the model does not take into account accidental factors such as cluster infections and super spenders under population movement.

III. THE ESTABLISHMENT OF POPULATION DENSITY MODEL

A. The Data Sources

The epidemic data of each region used in this paper are from the REAL-TIME dynamic DATABASE of COVID-19 epidemic of Dingxiang Yuan · Dingxiang Doctor^[3]. Draw a satellite map of the distribution area of the city based on the meet99 Network^[4]; The population density data of cities in a province in China are from Baidu database^[5]. The data about the population of cities in a province in China come from the

B. People Density Analysis

As the area where COVID-19 was detected earlier in China, a province in China has certain reference value in the study of novel coronavirus transmission. The province takes A city as the central city, which forms an urban circle with 8 surrounding cities. This urban circle is the most densely populated economic center of the province. As A central city, A City is also one of the regions with the largest population density in central China.

Compared with the SARS outbreak in 2003, the COVID-19 outbreak has shown a much more transmissible nature, and the spread of the epidemic has been greatly exacerbated by changes in population density brought about by population movements. Outbreaks in early stage of development, to coincide with the Chinese Spring Festival, large-scale, long distance migration caused local population density change quickly, transportation is the main place of the spread of the virus, the integration of population density parameters can more accurately calculate the spread of the virus in different parts of the speed, and provide appropriate defensive measures to solve the problem of different density and travel advice^[6].

In order to explore the transmission rate of novel coronavirus in different population densities, take cities in a province of China as an example and draw Figure 2 by analyzing the number of infected people, area and population in each region. The ratio of regional population density to infected people is calculated respectively (The population density is equal to the total population of the region divided by the area of the area. The proportion of infections is equal to the number of infected people in the region except the total population of the region.)

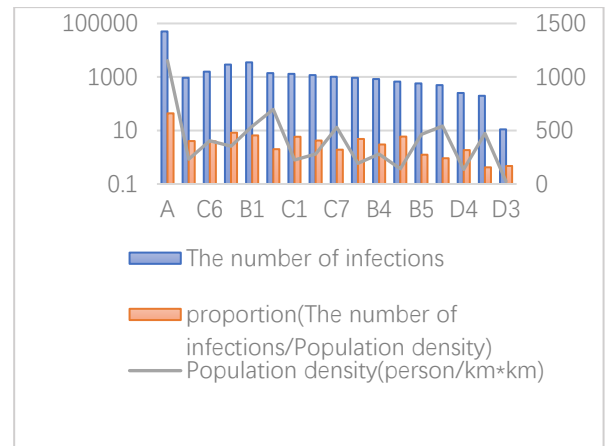


Fig. 2. An overview of infection rates in cities of a province in China

Analysis of the relationship between infection ratio and population density in each city shows that there is a certain relationship between epidemic spread and regional

population density. In this province, the epidemic spread to all sides from A City as the transmission center, and there was A high infection rate in the neighboring cities of A City, with A relatively large number of infections. In the same infected layer, the number of infected people increased rapidly in places with large population density. Where population density is low, the number of infections increases relatively slowly (note: the distance from the transmission center is related, and therefore has an impact on the relationship between some regions).

Figure 3 shows the data distribution map of cities in a province of China, when the population density started on the rise, spread infection range will jump to the second circle, and then has a similar rule, but the parts with a small population base, the characteristics of the area of an area small, the study of short duration.

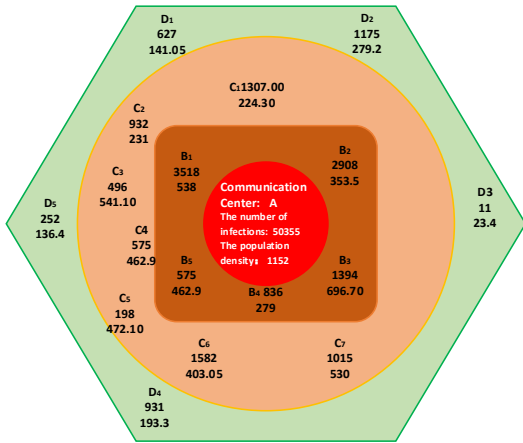


Fig. 3. Data distribution map of cities in a province of China

C. Virus Transmission Rate Analysis

Assuming that one person can infect n people in time t , the base number of infected people $i_0=1$, one infected person can infect 4 susceptible people.

First round of infection: $i(t)=1+1*4=5$;

Second round of infection: $i(t)=1+1*4+(1+1*4)*4=25$;

Third round of infection: $i(t)=1+1*4+(1+1*4)*4+(1+1*4+(1+1*4)*4)*4=125$

After n rounds of calculation, it can be seen that the total number of infected persons follows a geometric sequence, namely:

$$a_n = a_1 \cdot q^{(n-1)} \quad (4)$$

$$S_n = \frac{a_1(1-q^M)}{1-q} \quad (5)$$

Thus, the transmission speed is $v = i(t)/t$.

D. Model Building and Forecasting

1) *Population density model*: Considering the impact of population migration on the spread of the epidemic, the short-term prediction accuracy of the model may be insufficient if the constant infection rate is used to establish the model. Therefore, the study will consider the impact of population density on the transmission speed of the virus and establish the SIR model incorporating the parameter of "population density" [7].

Large-scale population movements are not conducive to population data collection and analysis. Therefore, the population is divided into different levels by region. Two population density cutting models are established to explore the impact of population density on the speed of transmission of the virus. There are two ways to study the spread of the virus in an area, one is the circular type (model A), and the other is the cutting type (model B). First, each data within the maximum range is obtained, and the number of times to be cut is calculated. Then, multiple sets of data are acquired through sequential division. Based on this, the changes in infected persons found under different population densities (the impact of the number of susceptible persons on the calculation should be considered) are integrated into the SIR model to calculate the speed of transmission at that population density. Compact circular models are separated from large to small areas, while cut models are separated from parallel areas. In this way, the population density and infection rate can be obtained to calculate the probability of transmission of the virus. According to the cutting model, a preliminary prediction can be made for the infection rate of the same virus under a fixed population density and the number of infected people in a period. Figure 4 shows that the spread of the virus in an area can be studied in two ways and presented in two cut models.

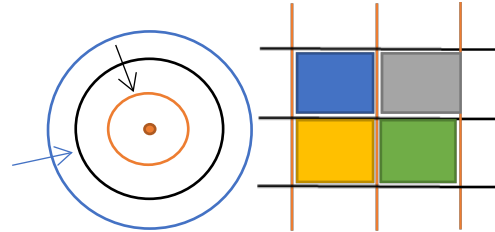


Fig. 4. Two cutting models

SIR model was used to predict the spread of the virus in Hubei Province [8, 9]. The data after different intervention policies were put into small areas, respectively, and the parameters λ and μ were obtained. Enter the parameters to calculate the actual transmission after intervention. Add a new parameter, v , divided by $s(t)$, $I(t)$, $r(t)$ of the corresponding region. To obtain the population density ratio of the area; In order to obtain the data under a certain population density.

$$\frac{s(t)}{v} = \frac{ds}{dtv} \quad \frac{i(t)}{v} = \frac{di}{dtv} \quad \frac{r(t)}{v} = \frac{dr}{dtv} \quad (6)$$

In the establishment of model A, the relationship between population density and infection rate was found by using the representative regions in A period of time, and the proportion of infected people to population density was also taken into account.

Infection reduction rate:

$$\frac{d\theta}{dt} = \frac{i_1}{i_2} \quad (7)$$

To obtain the reduction rate of infection multiplier at different levels:

$$\frac{dp}{dt} = \frac{p_1 - p_2}{p_1} \quad (8)$$

Error coefficient:

$$\frac{dk}{dt} = \theta - \beta \quad (9)$$

Population decline rate:

$$\frac{d\lambda}{dt} = s_1/s_2 \quad (10)$$

Figure 3 shows that the population density of central city A is about three times that of the diffusion area, and the number of infected people is also about three times that of the diffusion area, namely: $i_1/i_2=50355/17802=3.2499$. Population density ratio: $s_1/s_2=1152/344.2=2.9914$. Add error coefficient: $k=\theta-\lambda$. Then a formula can be obtained:

$$i_n = i_1 \cdot \left(\frac{d\lambda}{dt} + k\right)^{n-1} \quad (11)$$

The model can be used to predict the spread of infectious diseases in a certain region under the corresponding population density. Model B looks for other ways to integrate the infection relationship after drug intervention, and uses the Bayesian model to add data after other non-drug intervention to obtain the relationship between population density and infection rate of transmission. Taking a province in China as an example, the province is regarded as a large region with a population density of v and a small region of $v_1 v_2 v_3 v_4$.

2) *Intercity transmission model*: COVID-19 outbreak in a large city A, and comes as China travel peak, the height of the urban population of intensive and rapid flow, at the same time, wide range, large-scale migration flows between city people, lead to urban regional population density change extremely unstable, exacerbated by the spread of the virus speed, the spread of the epidemic there is A huge risk. In terms of external output, after the city was closed, the external traffic was immediately blocked. Compared with the population flow in the two weeks before the Spring Festival, the outflow of population from Wuhan to the whole country and Hubei province was significantly reduced in the two days after the city was closed, which controlled the spread of the epidemic to some extent.

a) *Space for virus transmission*: Correlation analysis showed that the virus spread from the central city as the starting point, and the central city as the radiation center, the main confirmed cases at the early stage of the outbreak is proportional to the inflow of the population in the central city, and to a certain extent, the impact of transportation and prevention and control measures. People in central cities migrate to surrounding areas in a certain period of time, and as the population density gradually increases, the number of infected cases also increases. The data on the spread of cases to surrounding cities were collected, and it was noted that the spread of the city itself and the impact of the spread of the surrounding cities on the city itself should be taken into account in the case of repeated cross-transmission, which would make the actual urban infection data higher than the original data.

In A city for typical cases, preliminary study the spread of CEMM model, is very obvious in infections or in the center of radiation to the highest strength, because when there is infection diffusion area, at the same time to spread of triple

area and radiation center, before the partition measures for high strength, the transmission will continue to increase, especially the periphery of internal secondary spread. Moreover, when the additional parameter population density was added, it was found that due to the change of regional area, the population density in the central region was much higher than that in other regions, thus increasing the transmission rate of the virus and increasing the number of infected people.

b) *Prediction*: According to the epidemic infection data of a province in China collected by Dxy, the map of the province was combined with the map information to analyze the location relationship among its cities and draw the map of level 4 infected areas in Figure 5.

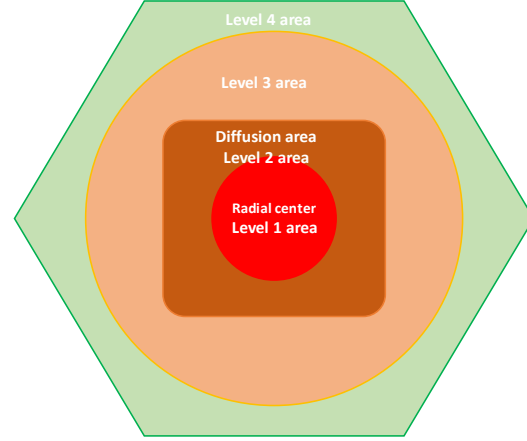


Fig. 5. Map of grade IV infected areas

In this paper, by studying CEMM propagation model analysis, taking City A as the radiation center and according to the geographical location of cities in the province, the data distribution map of cities in the province drawn in Figure 3 is integrated and classified, and the analysis diagram of virus spatial transmission in Figure 5 is obtained. Data in Figure 5 were used for analysis.

After the radiation center spread to the diffusion area, the number of infected cases in the diffusion area decreased by 2.825 times compared with the radiation center, and after the diffusion area spread to the tertiary area, the number of infected cases decreased by 3.049 times compared with the diffusion area. The infectious rate P between the two stages decreased by 7.9 ($\frac{d\rho}{dt} = \frac{p_1-p_2}{p_1}$).

Due to population flow, some infected people move across regions, which will have a certain impact on the primary model and generate multiple radiation centers in different regions. However, compared with the early spread of the epidemic in China, Wuhan is still the most typical radiation center, which has a relatively reliable reference value in certain regions. ρ : Reduction in infectious rate; θ : Infection rate; i_1 : Number of upper-level infections; i_2 : Next level of infected persons;

The study of this data model can be used to make a preliminary judgment on the diffusion area within a certain surrounding area from different radiation centers. However, if the cardinal number of this radiation center is much higher than the infection cardinal number outside the center, it is necessary to further study the diffusion data of other non-arbitrarily increased cardinal number areas.

However, the current research data can be used to study large-scale dispersal data from high-base areas worldwide. The above calculation data is for reference. When the radiation center spreads to the diffusion area, the population

density error coefficient k is used to determine the decay rate ρ of this area. Then, the lower transmittance is 7.929% lower than the upper region. Table I shows the visualized epidemic information.

TABLE I. VISUALIZED EPIDEMIC INFORMATION

city	Total population (1,000 people)	Number of infections (person)	population density (Person/km ²)	Proportion (number of infections/total population)
A	1121.20	50355	1152	0.449117%
C2	289.63	932	231.7	0.032179%
C6	557.01	1582	403.5	0.028402%
B2	633.30	2908	353	0.046225%
B1	492.10	3518	538.4	0.071490%
B3	105.97	1394	696.7	0.131547%
C1	222.10	1307	224.3	0.058847%
D2	568.00	1175	279.2	0.020687%
C7	268.93	1015	530	0.037742%
D4	413.79	931	193.3	0.022499%
B4	254.84	836	279.2	0.032805%
D1	339.80	672	141.5	0.019776%
B5	152.92	575	462.9	0.037601%
C3	124.74	496	541.1	0.039763%
D4	339.00	252	136.4	0.007434%
C5	96.61	198	472.1	0.020495%

E. Access to information

The real-time epidemic data of The Lilac Garden and the official website of the National Health Commission were obtained through the web crawler, given URL and rules, and the real-time confirmed cases, suspected cases, recovered cases and other data needed for prediction and analysis were obtained.

Send a request to the background through Ajax, and accept the returned JSON data, the Ajax technology for long connection process is: the front end sends a data request, the server receives the request to query whether there is new data, if there is no new data will block the request, until there is new data or timeout. The client sends a request immediately after receiving the result of the request. This method eliminates the step of JS timing request, and can get and return the information of new data in time. After the web page data is retrieved, the data is stored for further processing.

F. Information processing

The raw data format is collected unified, data cleaning is realized, and accurate data is obtained to acquire real-time epidemic data combined with modules such as rumor-refuting data and time. First, the python library is used for data analysis and processing, and a model framework is built. Node ordinal and various parameters (including infection rate, immunity rate, number of infected, number of immunes, number of nodes, and number of days) are set. Second, the library function SCIPy is used to determine the initial network node distribution and establish the calculation equation. The data parameter is introduced again. The Matplotlib library is used for visualization. Graphs of changes in the curves represented by the different data are plotted. Finally, a simulated social network is established. The obtained data is added to the social network as a parameter to simulate the spread of the virus and predict the result of the spread of the virus. Figure 6 reveals Ajax's long connection principle.

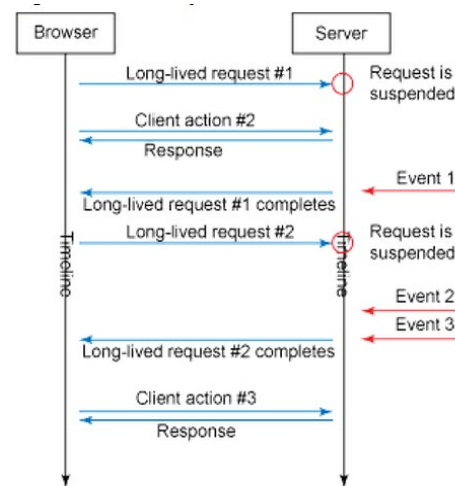


Fig. 6. Ajax long connection principle

G. Realization of information

The system background was built by Flask framework, and the epidemic information was visualized by Web front-end technology and Echarts chart. At the same time, the spatial distribution and temporal characteristics of the COVID-19 epidemic were visualized to achieve a good display effect. At the same time, SIR model and CEMM intercity model were improved to predict the epidemic situation.

Realize virus propagation simulation in social system. Through the Netlogo platform for the spread of the virus using abm method simulation modeling, complex system simulation, to intuitive understanding of the process the spread of the virus, the feature we by quoting SIR, SEIR, and other multiple transmission simulation model and using some simple data to show the communication effect, implement a simple propagation models show the spread of the process.

To achieve real-time analysis of the effect of epidemic

prevention measures and give effective epidemic prevention suggestions. The established prediction model includes Bayesian model, which can form different prediction results for different influencing factors. Therefore, epidemic prevention measures are taken as an important coefficient in calculation to simulate the epidemic transmission results. The coefficient is calculated by a comparison method in the model. The system compares various results, completes the screening, and returns the final results to the user, in order to complete and publish the most accurate and effective epidemic prevention bulletin.

IV. CONCLUSION

In this study, the traditional infectious disease model SIR model and CEMM model were analyzed and studied, and the "population density" parameter was integrated to improve the basic model and obtain a new prediction model, and the model evaluation and error analysis were carried out based on the epidemic data in China. The network crawler technology was used to crawl and summarize the COVID-19 epidemic information, and the data was cleaned. After sorting out the data format, the data was stored in a specific format as the basic data of this experiment, and the COVID-19 epidemic data set was obtained. The Web background and Echarts chart library were built to develop the epidemic information visualization system.

At the same time, the powerful Python library functions are used to realize the epidemic prediction simulation function, so that people can more intuitively understand the current situation of the epidemic, follow the situation of the epidemic anytime and anywhere, and obtain the epidemic information dynamically, so as to improve the awareness of protection. At the same time, the establishment of the epidemic information visualization system can make it easier for users to understand the mode and principle of the spread of the epidemic, and to master the epidemic situation in their region in advance. In addition, the platform also provides effective epidemic prevention plans and updates alert information to remind people to pay attention to epidemic

prevention and control and take protective measures.

The establishment of the prediction model of virus transmission and the realization of epidemic information visualization make full use of the advantages of the information age. The integration of mouth density parameters is beneficial to improve the reliability of epidemic prediction information. We studied the types and characteristics of information released by authoritative departments and institutions, strengthened the sharing of scientific and technological information in the fight against COVID-19, disseminated prevention and control knowledge to the public, and further improved their ability to perceive social risk information.

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